Memory-Efficient Torch Inference for AlphaGenome

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Abstract

We evaluated low-VRAM Torch inference strategies for AlphaGenome on a recreated human brain development ATAC checkpoint. At 131 kb and batch size 32, the lowest-memory configuration reduced observed peak memory from 90.46 GiB to 24.59 GiB, with test r changing from 0.80891 to 0.80860. At 1 Mb and batch size 2, default inference exceeded 48 GiB, whereas the complete fused configuration used 23.58 GiB.

1 Experimental setup

All experiments used the recreated merged Torch checkpoint from the best LoRA+LoCon finetuning run. We evaluated the human brain development ATAC head at resolution 128 on full valid and test splits. For 131 kb input windows, batch size was 32 and the evaluation contained 2162 examples. For 1 Mb input windows, a valid/test-only target cache was built, batch size was 2, and the evaluation contained 269 examples. Peak device memory was sampled with `nvidia-smi` and is reported as GiB. Accuracy is reported as test differential Pearson.

2 Implementation details

Bfloat16 policy. The `bf16_params` path uses the Torch model's aggressive bfloat16 dtype policy for inference. Floating-point model state and persistent activations are stored or cast to bfloat16 where supported by the Torch implementation, while integer inputs and indexing tensors are unchanged.

Effective standardized convolutions. AlphaGenome standardizes convolution weights at every forward pass. For an output channel, the effective inference weight is

$$W_{\text{eff}} = (W - \text{mean}(W))\,\text{rsqrt}\left(\max(\text{var}(W)\,\text{fan\_in}, 10^{-4})\right)\, s,$$

where s is the learned scale. The `_stdconv_effective` path materializes W_{eff} once and replaces the module with a direct convolution that preserves AlphaGenome SAME padding. This avoids runtime weight standardization and makes the layer eligible for the Triton Conv1d path.

Triton int8 weight-only Conv1d. Eligible wide Conv1d layers are replaced by a custom Triton kernel. Weights are stored as symmetric int8 values with one scale per output channel:

$$\alpha_c = \frac{\max|W_c|}{127}, \qquad Q_c = \text{round}(W_c/\alpha_c).$$

The kernel dequantizes inside the matrix multiply, accumulates in fp32 using `tl.dot`, and stores outputs in the input activation dtype. Heads, adapters, LoRA/LoCon modules, and other sensitive paths are excluded.

No-intermediates encoder path. When only 128 bp predictions are requested, the decoder does not consume the encoder U-Net skip tensors. The patched encoder therefore does not retain these intermediate tensors during evaluation.

Triton no-indices max-pooling. The encoder pool has kernel size 2 and stride 2. The custom Triton kernel loads the two candidate positions and stores only their maximum, matching the local ceil-length SAME pooling behavior while avoiding PyTorch's hidden max-pool index/workspace allocation.

Fused DNA embedder block. The `encoder.dna_embedder.block` module is fused into one Triton kernel implementing

$$\text{RMSBatchNorm} \rightarrow \text{GELU}_{\text{JAX}} \rightarrow \text{Conv1d}_{\text{int8 weight-only}}.$$

The RMSBatchNorm step uses stored running variance and affine parameters,

$$x_{\text{norm}} = x \cdot \gamma \cdot \text{rsqrt}(\text{running\_var} + \epsilon) + \beta.$$

The activation is AlphaGenome/JAX's sigmoid approximation rather than the standard erf GELU:

$$\text{GELU}_{\text{JAX}}(x) = x \cdot \sigma(1.702x).$$

The fused kernel applies normalization and activation immediately before the int8 weight-only convolution, without materializing the full-resolution normalized or activated tensors.

Fused first downsampling block. The `encoder.down_blocks.0` module is replaced by a wrapper containing two fused ConvBlocks and lean residual accumulation. In inference, the first block output already has the expanded channel dimension, so the residual input is added in-place into the leading input-channel slice instead of constructing an explicitly padded residual tensor. The second residual add is also in-place.

3 Results

Tables use feature indicators for bfloat16 policy (BF16), effective standardized convolutions (Eff.), Triton int8 Conv1d (Int8), no-intermediates encoder path (NoInt), Triton no-indices max-pool (Pool), fused DNA embedder block (FEmb), and fused first downsampling block (FD0).

Table 1: Full valid/test evaluation for 131 kb windows at batch size 32.

Strategy	BF16	Eff.	Int8	NoInt	Pool	FEmb	FD0	Peak GiB	Test r
Default								90.46	0.80891
Bfloat16	✓							65.37	0.80874
Triton Conv1d	✓	✓	✓					42.59	0.80877
No intermediates	✓	✓	✓	✓				42.59	0.80877
Triton pool	✓	✓	✓	✓	✓			34.09	0.80877
Fused embedder	✓	✓	✓	✓	✓	✓		31.59	0.80872
Fused down0	✓	✓	✓	✓	✓		✓	30.59	0.80860
Fused embedder+down0	✓	✓	✓	✓	✓	✓	✓	24.59	0.80860

At 131 kb, the lowest-memory successful configuration was the fused embedder+down0 strategy, reducing observed peak memory from 90.46 GiB to 24.59 GiB relative to default. The best non-fused memory-efficient option was the Triton-pool strategy, at 34.09 GiB with no visible test metric change relative to the Triton Conv1d baseline.

Table 2: Full valid/test evaluation for 1 Mb windows at batch size 2.

Strategy	BF16	Eff.	Int8	NoInt	Pool	FEmb	FD0	Peak GiB	Test r
Default								51.36	0.82707
Bfloat16	✓							46.48	0.82640
Triton Conv1d	✓	✓	✓					32.58	0.82662
No intermediates	✓	✓	✓	✓				28.58	0.82662
Triton pool	✓	✓	✓	✓	✓			30.08	0.82662
Fused embedder	✓	✓	✓	✓	✓	✓		28.83	0.82658
Fused down0	✓	✓	✓	✓	✓		✓	26.58	0.82643
Fused embedder+down0	✓	✓	✓	✓	✓	✓	✓	23.58	0.82634

At 1 Mb, default inference exceeded 48 GiB. Plain bfloat16 inference fit just under 48 GiB, while the Triton/no-intermediates family reduced memory substantially further. The fused blocks primarily reduced observed memory at this longer context length; they are useful for minimum VRAM but not for maximum throughput.

4 Discussion

The most robust practical configuration is the combination of bfloat16, precomputed standardized convolutions, Triton int8 weight-only Conv1d, the 128 bp no-intermediates encoder path, and Triton no-indices max-pooling. This configuration kept peak memory below 48 GiB for both tested input lengths while preserving test differential Pearson relative to the Triton Conv1d baseline. The fused high-resolution blocks are useful when minimizing observed memory is more important than maximizing throughput.